SUSTech Southern University of Science and Technology

SME206, Signals and Systems

Lab3 – Fourier Series of Periodic Signals

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Outline

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- Tutorial: fft, ifft
- Synthesizing Signals with Discrete-Time Fourier Series
- Synthesizing Continuous-Time signals with the Fourier Series
- The Fourier Representation of Square Waves



1、Tutorial: fft, ifft

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Background: Discrete-Time Fourier Series (DTFS)

$$x[n] = \sum_{k=\langle N \rangle} a_k \phi_k[n] = \sum_{k=\langle N \rangle} a_k e^{jk\left(\frac{2\pi}{N}\right)n}$$

$$a_k = \langle x[n], \phi_k[n] \rangle = \frac{1}{N} \sum_{n=\langle N \rangle} x[n]\phi_k^*[n] = \frac{1}{N} \sum_{n=\langle N \rangle} x[n] e^{-jk\left(\frac{2\pi}{N}\right)n}$$

Note:

- N: fundamental period

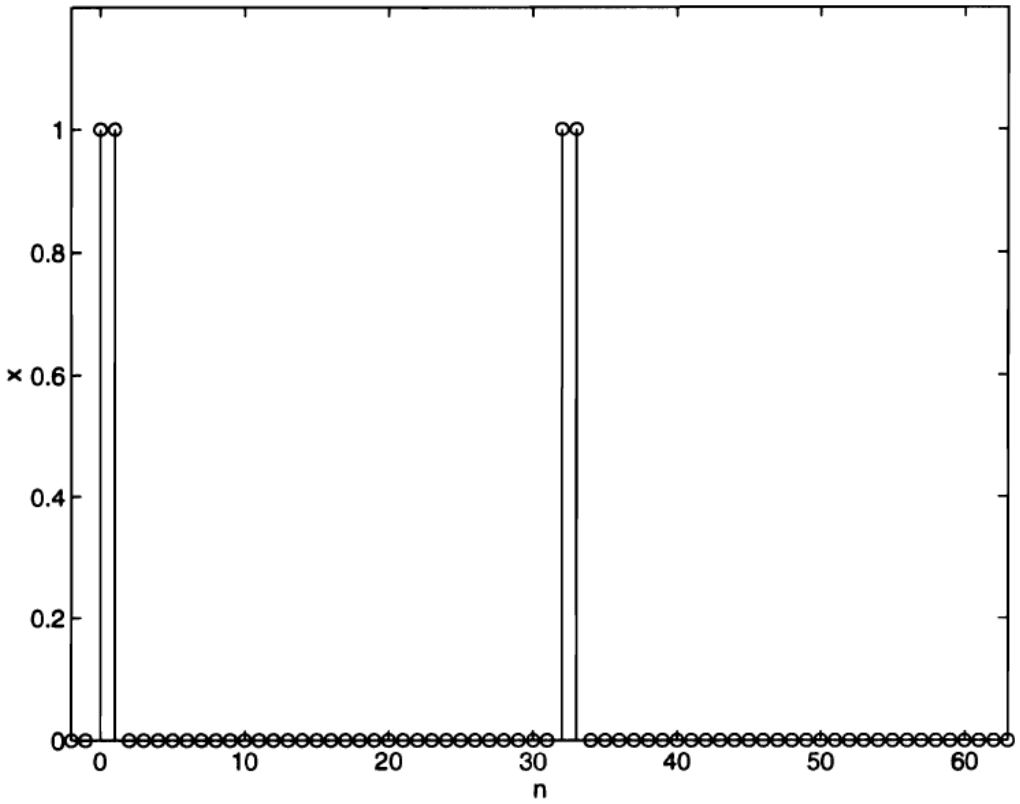
  <N>: any N consecutive points period, e.g [0, 1, 2, ⋯, N-1]

- a_k : DTFS coefficients

1、Tutorial: fft, ifft

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DTFS Analysis, Calculation step by step



$$x[n] = \begin{cases} 1, & n = 0, 1, \\ 0, & \text{otherwise}. \end{cases}$$

Figure 3.1. Periodic discrete-time signal $x[n]$ with fundamental period $N = 32$.

1、Tutorial: fft, ifft

N = 32

$$x[n] = \begin{cases} 1, & n = 0, 1, \\ 0, & \text{otherwise}. \end{cases}$$

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DTFS Analysis, Calculation step by step

$$a_k = \frac{1}{N}\sum_{n=0}^{N-1} x[n]e^{-jk\frac{2\pi}{N}n}$$

$$a_0 = \frac{1}{32}\sum_{n=0}^{N-1} x[n]e^{-j0\frac{2\pi}{32}n} = \frac{1}{32}\sum_{n=0}^{N-1} x[n] = \frac{1}{32}(x[0] + x[1]) = \frac{1}{16} = 0.0625$$ --- Average / DC part

$$a_1 = \frac{1}{32}\sum_{n=0}^{N-1} x[n]e^{-j1\frac{2\pi}{32}n} = \frac{1}{32}\left(x[0] + x[1]e^{-j\frac{\pi}{16}}\right) = \frac{1}{32}\left(1 + e^{-j\frac{\pi}{16}}\right) =$$

```
>> 1/32 * (1 + exp(-j * pi / 16) )

ans =

  0.0619 - 0.0061i
```

.....

$$a_{31} = \frac{1}{32}\sum_{n=0}^{N-1} x[n]e^{-j31\frac{2\pi}{32}n} = \frac{1}{32}\left(x[0] + x[1]e^{-j\frac{31\pi}{16}}\right) = \frac{1}{32}\left(1 + e^{-j\frac{31\pi}{16}}\right) =$$

```
>> 1/32 * (1 + exp(-j * 31 * pi / 16) )

ans =

  0.0619 + 0.0061i
```

1、Tutorial: fft, ifft

N = 32

$$x[n] = \begin{cases} 1, & n = 0, 1, \\ 0, & \text{otherwise}. \end{cases}$$

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DTFS Analysis, Calculation by 'fft' function

$$a_k = \frac{1}{N} \sum_{n=0}^{N-1} x[n] e^{-jk\frac{2\pi}{N}n}$$

```
N = 32;
n = 0:(N-1);
x = 1.0 *(n ==0) + 1.0*(n==1);

a = (1/N) * fft(x);
```

Easy Usage

=> input signals X in whole fundamental period

=> final result need to divided by N

1x32 complex double

	a_0 1	a_1 2	3	.....	31	a_{31} 32
1	0.0625 + 0.0000i	0.0619 - 0.0061i	0.0601 - 0...		0.0601 + ...	0.0619 + 0.0061i

1、Tutorial: fft, ifft

DTFS Analysis

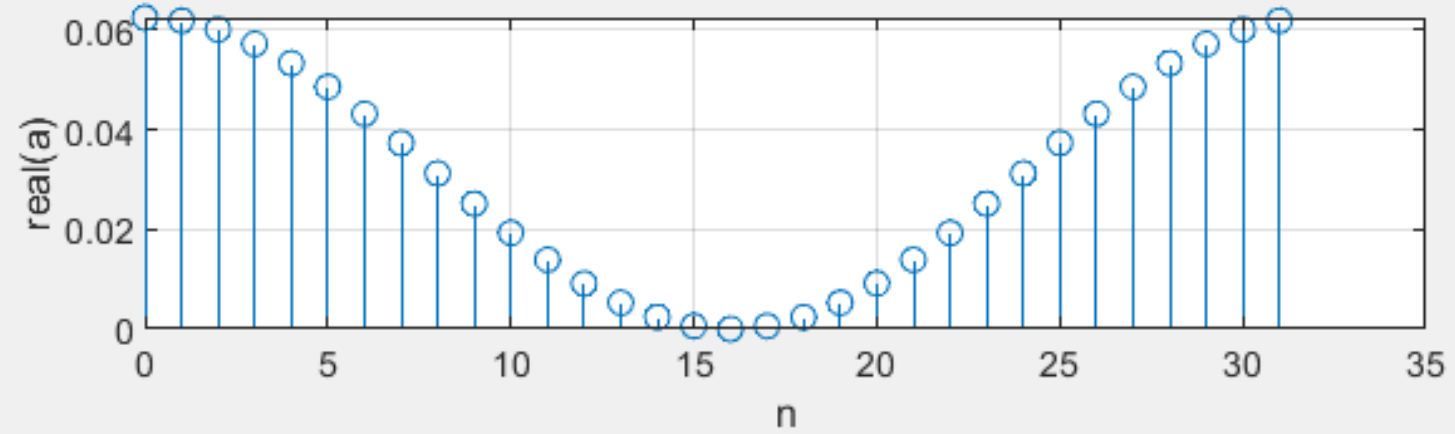
N = 32

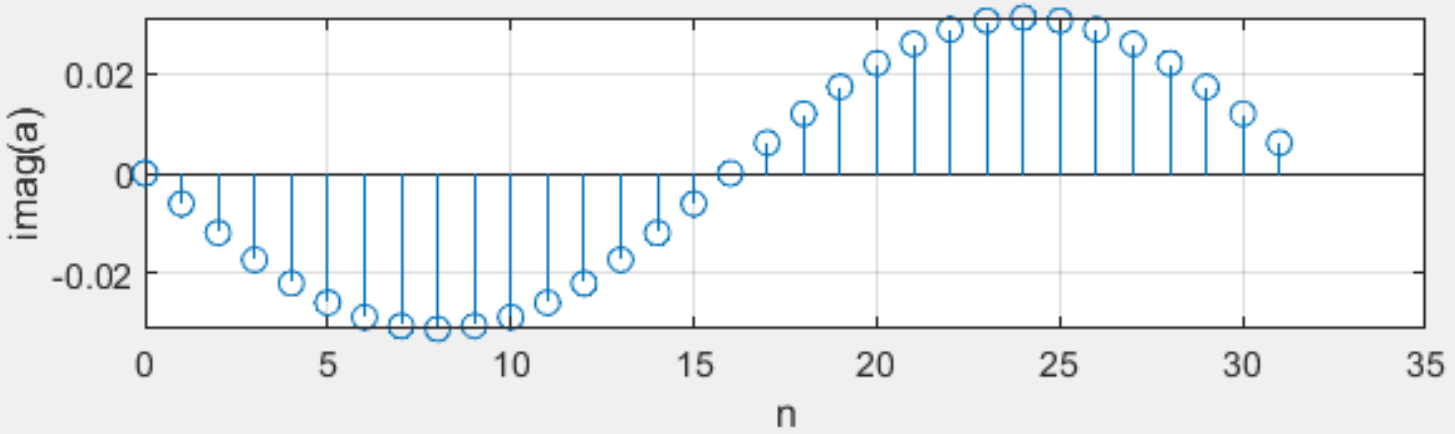
$$x[n] = \begin{cases} 1, & n = 0, 1, \\ 0, & \text{otherwise}. \end{cases}$$

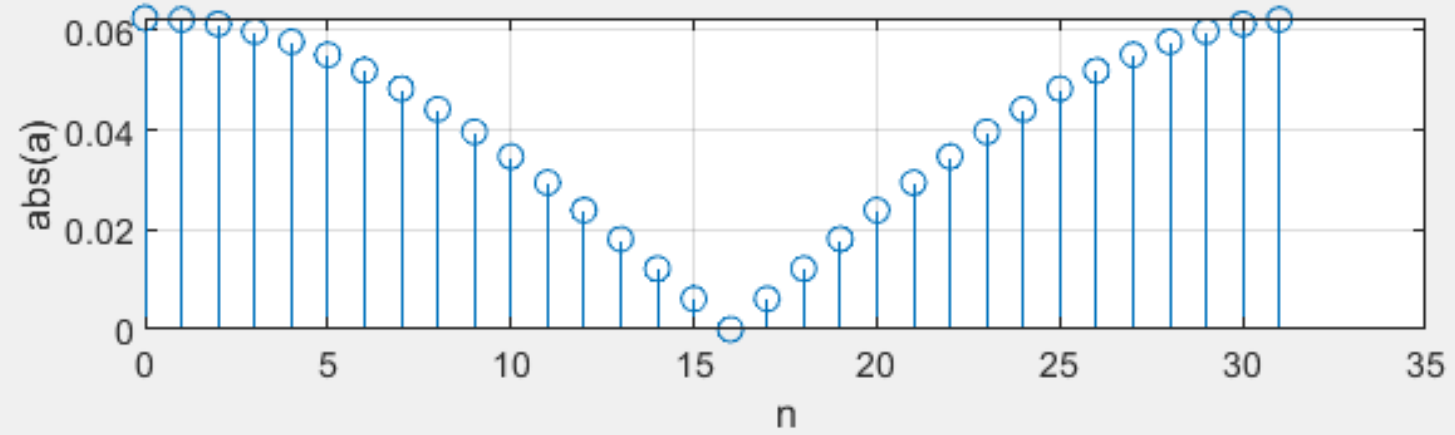
```
N = 32;
n = 0:(N-1);
x = 1.0 *(n ==0) + 1.0*(n==1);
```

$$a_k = \frac{1}{N} \sum_{n=0}^{N-1} x[n] e^{-jk\frac{2\pi}{N}n}$$

```
a = (1/N) * fft(x);
```







1、Tutorial: fft, ifft

N = 32

$$x[n] = \begin{cases} 1, & n = 0, 1, \\ 0, & \text{otherwise}. \end{cases}$$

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DTFS Synthesis, Calculation by sum

$$x[n] = \sum_{k=0}^{N-1} a_k e^{jk\frac{2\pi}{N}n}$$

$$x[0] = \sum_{k=0}^{N-1} a_k e^{jk\frac{2\pi}{32}0} = \sum_{k=0}^{N-1} a_k = 1$$

$$x[1] = \sum_{k=0}^{N-1} a_k e^{jk\frac{2\pi}{32}1} = \sum_{k=0}^{N-1} a_k e^{jk\frac{\pi}{16}} = 1$$

```
x_systhesis = zeros(1, N);
for n = 0:(N-1)
    for k = 0:(N-1)
        x_systhesis(n+1) = x_systhesis(n+1) + a(k+1) * exp(j * k * 2 * pi / 32 * n);
    end
end
```

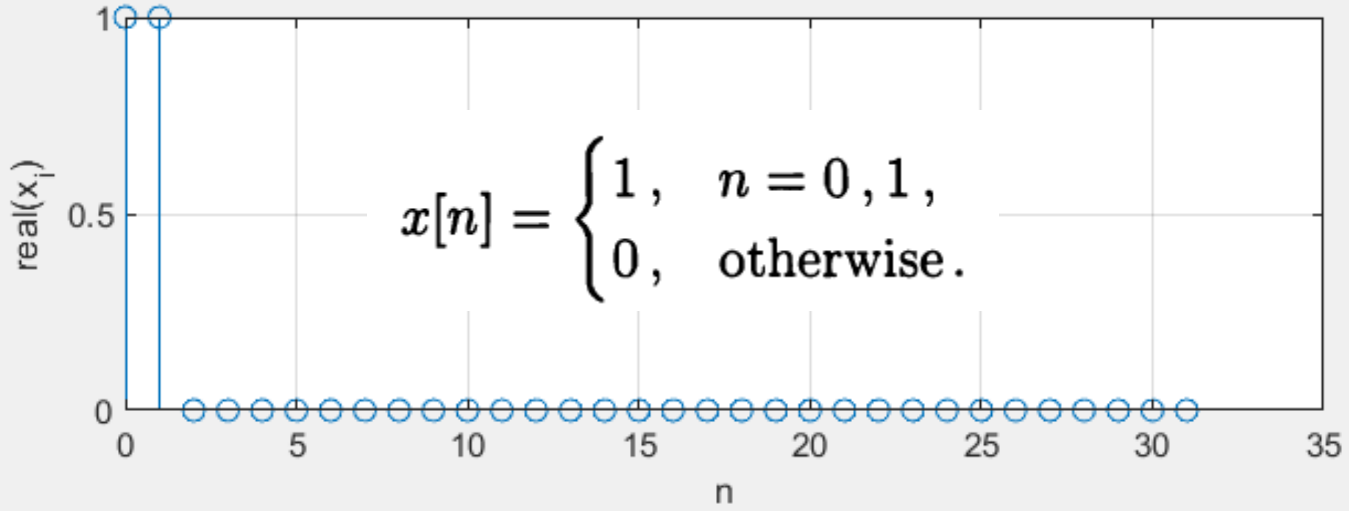
1、Tutorial: fft, ifft

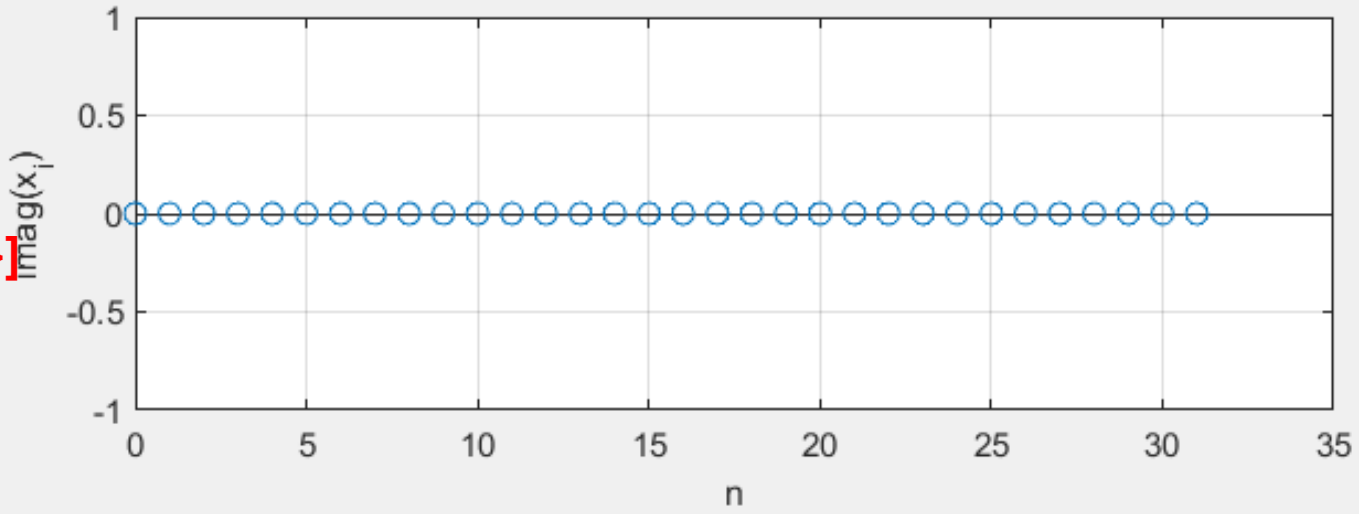
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DTFS Synthesis, Calculation by "ifft" function

$$x[n] = \sum_{k=0}^{N-1} a_k e^{jk\frac{2\pi}{N}n}$$

```
% x_i = ifft(N * a);
x_i = N * ifft(a);
```





Easy Usage

=> input DTFS coefficients, [a0, a1, … a{N-1}]

=> final result multiply by N, [x0, x1, … x{N-1}]

2, Synthesizing Signals with Discrete-Time Fourier Series

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From Lab Book 3.5), as Lab Assignment

- Basic: synthesis a simple periodic DT signal from DTFS coefficients
- Intermediate: analyze DT signal to get DTFS coefficients, and construct back
- Advanced: about DTFS computation details



2, Synthesizing Signals with Discrete-Time Fourier Series

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Basic: synthesis a simple periodic DT signal from DTFS coefficients

In these problems, you will synthesize a periodic discrete-time signal with period $N = 5$ and the following DTFS coefficients

$$a_0 = 1,\ a_2 = a_{-2}^* = e^{j\pi/4},\ a_4 = a_{-4}^* = 2e^{j\pi/3}.$$

(a). Based on the DTFS coefficients, do you expect $x[n]$ to be complex-valued, purely real, or purely imaginary? Why?

(b). Using the DTFS coefficients given above, determine the values of a_0 through a_4 and specify a vector `a` containing these values.

(c). Using the vector `a` of DTFS coefficients and the synthesis equation, define a new vector `x` containing one period of the signal $x[n]$ for $0 \le n \le 4$. You can either write out the summation explicitly or you may find it helpful to use a `for` loop. Generate an appropriately labeled plot of $x[n]$ for $0 \le n \le 4$ using `stem`. Was your prediction in Part (a) correct? Note that if you predicted a purely imaginary or real signal, it may still have a very small ($< 10^{-10}$) nonzero real or imaginary part due to roundoff errors. If this is the case, set this part to be zero using `real` or `imag` as appropriate before making your plot.

a) Signal Analysis: real or imaginary?

b) Properties of DTFS coefficients?

c) Signal synthesis by 'ifft'

More: DTFS coefficients

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a_k is periodic

$$a_{rN+k} = a_k, whre\ r = \pm 1, \pm 2, \ldots$$

=> Proof

$$a_{rN+k} = \frac{1}{N}\sum_{n=0}^{N-1} x[n]e^{-j(rN+k)\frac{2\pi}{N}n} = \frac{1}{N}\sum_{n=0}^{N-1} x[n]e^{-j2\pi rn}e^{-jk\frac{2\pi}{N}n}$$

$$= \frac{1}{N}\sum_{n=0}^{N-1} x[n]e^{-jk\frac{2\pi}{N}n} = a_k$$

More: DTFS coefficients

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a_k is periodic

$$a_{rN+k} = a_k, where\ r = \pm 1, \pm 2, \ldots$$

$$\Rightarrow \quad a_{N-k} = a_{-k}$$

For the signal x[n] with period of N = 5, DTFS coefficients:

Form #1: $[a_0, a_1, a_2, a_3, a_4]$

$$a_3 = a_{-2}$$
$$a_4 = a_{-1}$$

For fft, ifft function

Form #2: $[a_{-2}, a_{-1}, a_0, a_1, a_2]$

For Harmonic wave

$$x[n] = a_0 + \left(a_1 e^{j\frac{2\pi}{N}n} + a_{-1} e^{-j\frac{2\pi}{N}n}\right) + \left(a_2 e^{j2\frac{2\pi}{N}n} + a_{-2} e^{-j2\frac{2\pi}{N}n}\right)$$

DC component

1st harmonic component

2nd harmonic component

More: DTFS coefficients

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When x[n] is real signal => $a_k = a^*_{-k}$

For the signal x[n] with period of N = 5:

$$x[n] = a_0 + \underbrace{\left(a_1 e^{j\frac{2\pi}{N}n} + a_{-1} e^{-j\frac{2\pi}{N}n}\right)}_{\text{real}} + \underbrace{\left(a_2 e^{j2\frac{2\pi}{N}n} + a_{-2} e^{-j2\frac{2\pi}{N}n}\right)}_{\text{real}}$$

2, Synthesizing Signals with Discrete-Time Fourier Series

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Intermediate: analyze DT signal to get DTFS coefficients, and construct back

For these problems, you will examine the DTFS representation of several different square waves. Specifically, you will look at signals

$$x_1[n] = \begin{cases} 1, & 0 \le n \le 7, \end{cases} \tag{3.4}$$

$$x_2[n] = \begin{cases} 1, & 0 \le n \le 7, \\ 0, & 8 \le n \le 15, \end{cases} \tag{3.5}$$

Assignment

$$x_3[n] = \begin{cases} 1, & 0 \le n \le 7, \\ 0, & 8 \le n \le 31, \end{cases} \tag{3.6}$$

where $x_1[n]$, $x_2[n]$ and $x_3[n]$ have periods of $N_1 = 8$, $N_2 = 16$ and $N_3 = 32$, respectively.

2, Synthesizing Signals with Discrete-Time Fourier Series

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Intermediate: analyze DT signal to get DTFS coefficients, and construct back

(d). Define three vectors `x1`, `x2`, and `x3` representing one period of each of the signals $x_1[n]$, $x_2[n]$ and $x_3[n]$. Using these vectors, make appropriately labeled plots of each of the signals over the range $0 \leq n \leq 63$. Note: You will have to repeat the vectors you have defined to cover this range of samples.

d) Define the signals, and plot

(e). Exercise 3.1 explains how to use `fft` to compute the DTFS of a periodic discrete-time signal from one period of the signal. Using the `fft` function, define the vectors `a1`, `a2`, and `a3` to be the DTFS coefficients of $x_1[n]$ through $x_3[n]$, respectively. Generate appropriately labeled plots of the magnitude of each of the DTFS coefficient sequences using `abs` and `stem`. From your time domain plots of Part (d) and Eq. (3.2), you should be able to predict the values of `a1(1)`, `a2(1)` and `a3(1)` — the DC components of the signals. Do your predicted values match those obtained with MATLAB?

e) Calculate DTFS coefficients , and its distribution diagram

2, Synthesizing Signals with Discrete-Time Fourier Series

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Intermediate: analyze DT signal to get DTFS coefficients, and construct back

(f). In this part, you will examine how $x_3[n]$ appears when it is synthesized a few coefficients at a time. Using the vector `a3` you found in the previous part, define vectors `x3_2`, `x3_8`, `x3_12` and `x3_all` corresponding to the following four signals

$$x_{3_2}[n] = \sum_{k=-2}^{2} a_k e^{jk(2\pi/32)n},$$

$$x_{3_8}[n] = \sum_{k=-8}^{8} a_k e^{jk(2\pi/32)n},$$

$$x_{3_12}[n] = \sum_{k=-12}^{12} a_k e^{jk(2\pi/32)n},$$

$$x_{3_all}[n] = \sum_{k=-15}^{16} a_k e^{jk(2\pi/32)n},$$

on the interval $0 \le n \le 31$. Note that since $x_3[n]$ is real, the DTFS coefficients for this signal will be conjugate symmetric, i.e., $a_k = a^*_{-k}$. Because a_k is conjugate symmetric and all of the sums except $x_{3_all}[n]$ are symmetric about $k = 0$, the resulting time signals should be purely real. If you are unclear about why this is true, you may want to review the symmetry properties of the DTFS. Due to roundoff error in MATLAB, you may need to discard some very small imaginary parts of the signals you synthesize using `real`. The sums specified above are symmetric about $k = 0$ but the vector `a3` represents a_k for $k = 0, \ldots, 31$ as `[a3(1),...,a3(32)]`, so you will need to determine which elements of `a` correspond to the negative values of k when you implement the sums.

(g). Argue that $x_{3_all}[n]$ must be a real signal.

(h). Generate a sequence of appropriately labeled plots using `stem` showing how the signals you created converge to $x_3[n]$ as more of the DTFS coefficients are included in the sum. Specifically, `x3_all` should be equal to the original vector `x3` within the bounds of MATLAB's roundoff error. Does the synthesis of this discrete-time square wave display the Gibb's phenomenon?

Signal synthesis by 'ifft' : 0~n harmonics

=> Is the signal gradually approximating?

=> Gibb's problem is optional

Lab3 / Task

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3.5:

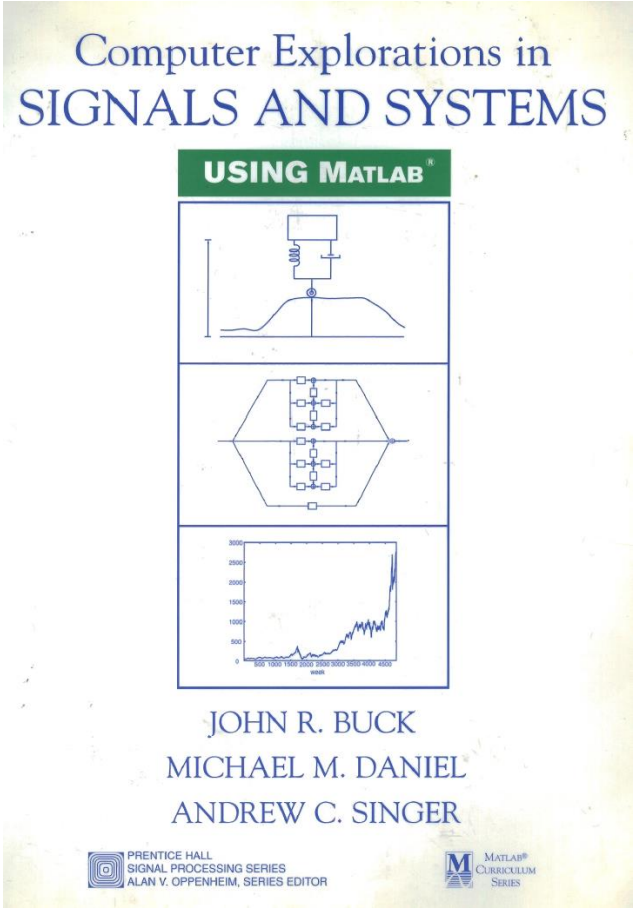
- a) ~ c)
- d) ~ h) for x3
- Except Gibb ‘s problem of h)

3.12:

- a) ~ d), for square wave

Deadline: see the end

Submit to Blackboard



Background: CTFS

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Continuous-Time Fourier Series (CTFS)

$$x(t) = \sum_{k=-\infty}^{+\infty} a_k \phi_k(t) = \sum_{k=-\infty}^{+\infty} a_k e^{jk(2\pi/T)t}$$

$$a_k = \langle x(t), \phi_k(t) \rangle = \frac{1}{T} \int_T x(t) \phi_k^*(t) dt$$

$$= \frac{1}{T} \int_T x(t) e^{-jk(2\pi/T)t} dt$$

Note:

- T: fundamental period. eg, [0, T]
- a_k : CTFS coefficients

Background: CTFS

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For CTFS, a_k is periodic ?

For DTFS, a_k is periodic

$$a_{rN+k} = a_k, where\ r = \pm 1, \pm 2, \ldots$$

$$a_{rN+k} = \frac{1}{N}\sum_{n=0}^{N-1} x[n]e^{-j(rN+k)\frac{2\pi}{N}n} = \frac{1}{N}\sum_{n=0}^{N-1} x[n]e^{-j2\pi rn}e^{-jk\frac{2\pi}{N}n}$$

$$= \frac{1}{N}\sum_{n=0}^{N-1} x[n]e^{-jk\frac{2\pi}{N}n} = a_k$$

Background: CTFS

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For CTFS, a_k is periodic ? ➔ NO

=> Assuming the existence of period N

$$a_{k+rN} = \frac{1}{T}\int_0^T x(t)\, e^{-j(k+rN)\frac{2\pi}{T}t}dt = \frac{1}{T}\int_0^T x(t)\, e^{-jk\frac{2\pi}{T}t}\, e^{-jrN\frac{2\pi}{T}t}dt$$

CAN NOT be eliminated

=> Additionally, from Parseval's Relation

For CTFS $$\frac{1}{T}\int_T |x(t)|^2 dt = \sum_{k=-\infty}^{+\infty} |a_k|^2$$

Due to periodicity, and there must be a term a_k is not 0,

So is the accumulation of this cycle infinite ?

For DTFS, a_k is periodic

$$a_{rN+k} = a_k, where\ r = \pm 1, \pm 2, \ldots$$

$$a_{rN+k} = \frac{1}{N}\sum_{n=0}^{N-1} x[n]e^{-j(rN+k)\frac{2\pi}{N}n} = \frac{1}{N}\sum_{n=0}^{N-1} x[n]e^{-j2\pi rn}e^{-jk\frac{2\pi}{N}n}$$

Eliminated

$$= \frac{1}{N}\sum_{n=0}^{N-1} x[n]e^{-jk\frac{2\pi}{N}n} = a_k$$

For DTFS $$\frac{1}{N}\sum_{n=<N>} |x[n]|^2 = \sum_{k=<N>} |a_k|^2$$

Background: CTFS

Showcase, From Lab book 3.12) / Triangle Wave

Analytically find the coefficients of the CTFS for triangle wave…

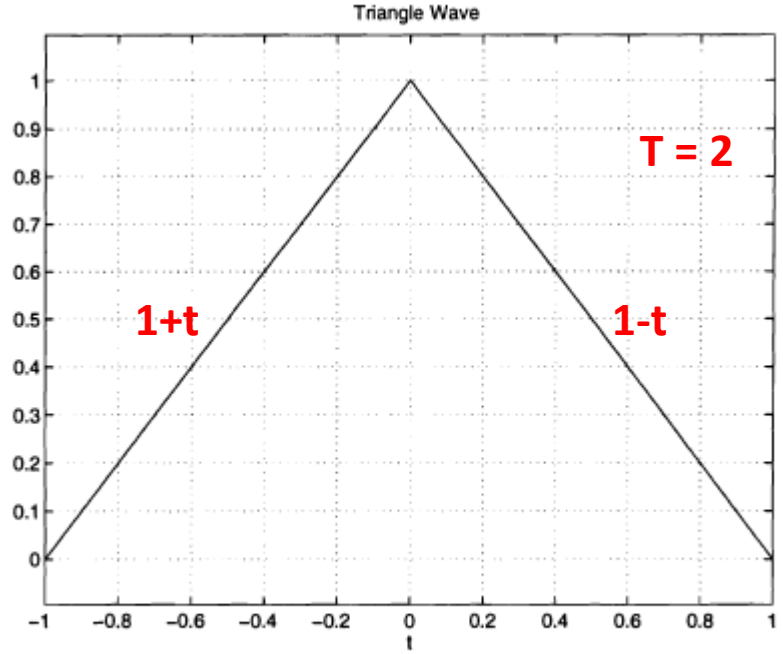


Figure 3.7. A single period of the triangle wave.

$$a_k = \frac{1}{T}\int_0^T x(t)\, e^{-jk\frac{2\pi}{T}t}dt = \int_{-1}^{0}(1+t)e^{-jk\pi t}dt + \int_0^1 (1-t)e^{-jk\pi t}dt$$

$$= \int_{-1}^{1} e^{-jk\pi t}dt + \int_{-1}^{0} t\, e^{-jk\pi t}dt - \int_0^1 t\, e^{-jk\pi t}dt$$

$$= \frac{j}{k\pi}e^{-jk\pi t}\Big|_{-1}^{1} + \frac{j}{k\pi}t\, e^{-jk\pi t}\Big|_{-1}^{0} + \frac{1}{(k\pi)^2}e^{-jk\pi t}\Big|_{-1}^{0} - \frac{j}{k\pi}t\, e^{-jk\pi t}\Big|_{0}^{1} - \frac{1}{(k\pi)^2}e^{-jk\pi t}\Big|_{0}^{1}$$

$$= \frac{j}{k\pi}e^{-jk\pi} - \frac{j}{k\pi}e^{jk\pi} + \frac{j}{k\pi}e^{jk\pi} + \frac{1}{(k\pi)^2}\left(1-e^{jk\pi}\right) - \frac{j}{k\pi}e^{-jk\pi} + \frac{1}{(k\pi)^2}\left(1-e^{-jk\pi}\right)$$

$$= \frac{2}{(k\pi)^2}(1-\cos(k\pi)) = \frac{4\sin^2\left(\frac{k\pi}{2}\right)}{(k\pi)^2} = \left(\frac{\sin\left(\frac{k\pi}{2}\right)}{\frac{k\pi}{2}}\right)^2$$

Background: CTFS

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Showcase, From Lab book 3.12) / Triangle Wave

Matlab calculate the coefficients of the CTFS for triangle wave…

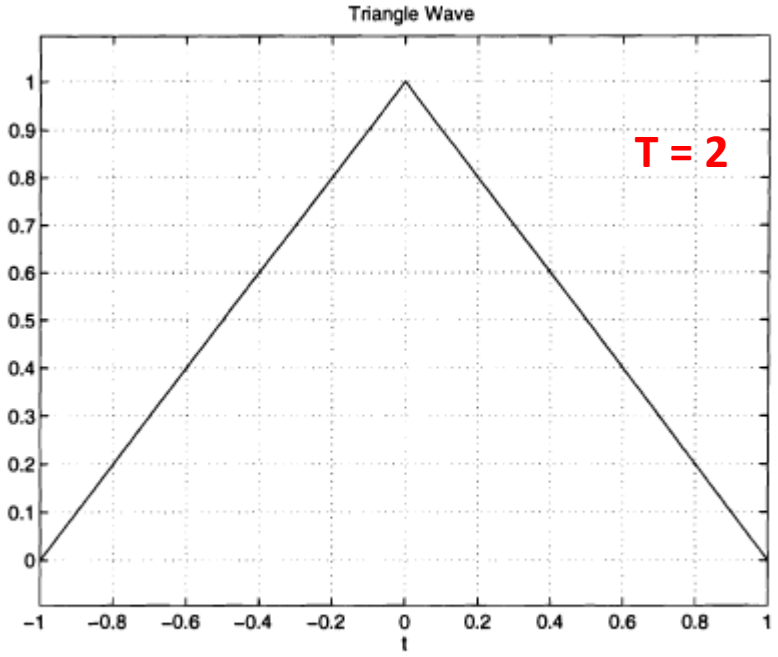


Figure 3.7. A single period of the triangle wave.

```
syms t;
syms k;

xt = 1 - abs(t);
T = 2;

ak = 1 / T * int( xt * exp(-j * k * (2*pi/T) *
t), t, [-T/2, T/2]);

a0 = 1 / T * int( xt, t, [-T/2, T/2]);
```

Background: CTFS

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Showcase, From Lab book 3.12) / Triangle Wave

Matlab calculate the coefficients of the CTFS for triangle wave…

$$\left(\frac{\sin\left(\frac{k\pi}{2}\right)}{\frac{k\pi}{2}}\right)^2$$

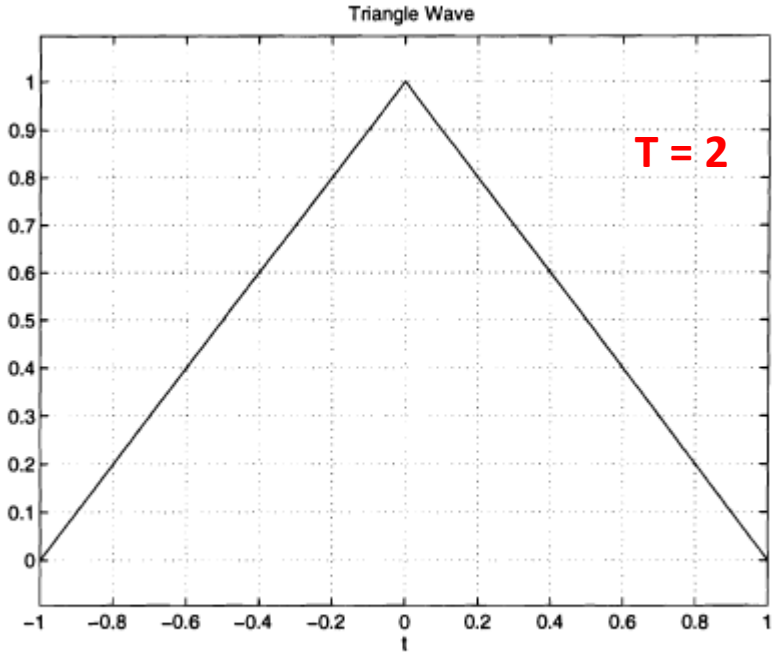


Figure 3.7. A single period of the triangle wave.

After internal calculation in Matlab

```
syms t;
syms k;

xt = 1 - abs(t);
T = 2;

ak = 1 / T * int( xt * exp(-j * k * (2*pi/T) * t), t, [-T/2, T/2]);

a0 = 1 / T * int( xt, t, [-T/2, T/2]);
```

Background: CTFS

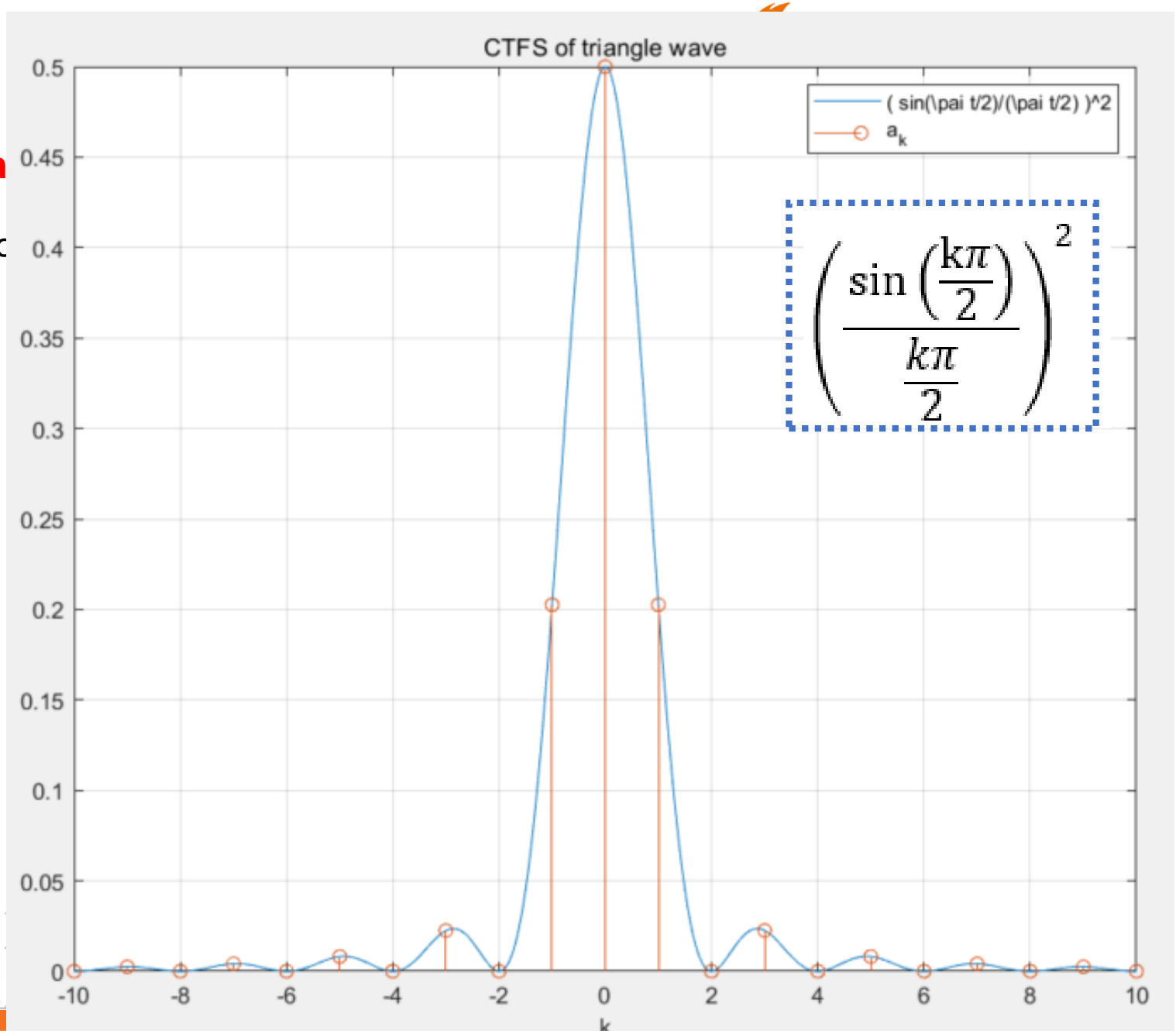
Showcase, From Lab book 3.12) / Tria

Matlab calculate the coefficients of the CTFS fc

```
ezplot(ak, [-10, 10]);
hold on;

k10 = [-10:-1, 1:10];
a_k10 = subs(ak, k, k10);
a_k10_val = double(a_k10);

stem([k10, 0], [a_k10_val, a0]);
```



Background: CTFS

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Showcase, From Lab book 3.6)

$$x_1(t) = \cos(\omega_0 t) + \sin(2\omega_0 t), \qquad \text{where } \omega_0 = 2\pi.$$

Analytically find the coefficients of the CTFS for x1(t)

⇨ T = 1

$$a_k = \frac{1}{T}\int_0^T x_1(t)\, e^{-jk\frac{2\pi}{T}t} dt = \int_0^1 \left[\frac{1}{2}\left(e^{j2\pi t} + e^{-j2\pi t}\right) - \frac{j}{2}\left(e^{j4\pi t} - e^{-j4\pi t}\right)\right] e^{-jk2\pi t} dt$$

$$= \frac{1}{2}\int_0^1 e^{-j(k-1)2\pi t}\, dt + \frac{1}{2}\int_0^0 e^{-j(k+1)2\pi t}\, dt - \frac{j}{2}\int_0^0 e^{-j(k-2)2\pi t}\, dt + \frac{j}{2}\int_0^0 e^{-j(k+2)2\pi t}\, dt$$

..... Better Way than this ??

Background: CTFS

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Showcase, From Lab book 3.6)

$$x_1(t) = \cos(\omega_0 t) + \sin(2\omega_0 t), \qquad \text{where } \omega_0 = 2\pi.$$

$$x(t) = \sum_{k=-\infty}^{+\infty} a_k \phi_k(t) = \sum_{k=-\infty}^{+\infty} a_k e^{jk(2\pi/T)t}$$

Back to basic

Analytically find the coefficients of the CTFS for x1(t)

$$x_1(t) = \cos(2\pi t) + \sin(4\pi t) = \frac{1}{2}\left(e^{j2\pi t} + e^{-j2\pi t}\right) - \frac{j}{2}\left(e^{j4\pi t} - e^{-j4\pi t}\right)$$

$$= \frac{1}{2}e^{j\frac{2\pi}{1}t} + \frac{1}{2}e^{j(-1)\frac{2\pi}{1}t} - \frac{j}{2}e^{j2\frac{2\pi}{1}t} + \frac{j}{2}e^{j(-2)\frac{2\pi}{1}t})$$

a_1 a_{-1} a_2 a_{-2}

$others = 0$

Background: Properties of CTFS

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Showcase, From Lab book 3.6)

$$x_1(t) = \cos(\omega_0 t) + \sin(2\omega_0 t), \qquad \text{where } \omega_0 = 2\pi.$$

(b). Consider the signal $y(t) = x_1(t) + x_1(-t)$. Using the time-reversal and conjugation properties of the CTFS, determine the coefficients for the CTFS of $y(t)$.

(c). Plot the signal $y(t)$ over $-1 \le t \le 1$. What type of symmetry do you expect? Can you explain what you see in terms of the symmetry properties of the CTFS?

Linearity

$$x(t) \overset{FS}{\longleftrightarrow} a_k$$

$$y(t) \overset{FS}{\longleftrightarrow} b_k$$

$$z(t) = Ax(t) + By(t) \overset{FS}{\longleftrightarrow} c_k = Aa_k + Bb_k$$

Time Reversal

$$x(t) \overset{FS}{\longleftrightarrow} a_k$$

$$x(-t) \overset{FS}{\longleftrightarrow} a_{-k}$$

Background: Properties of CTFS

Showcase, From Lab book 3.6)

$$x_1(t) = \cos(\omega_0 t) + \sin(2\omega_0 t), \qquad \text{where } \omega_0 = 2\pi.$$

(b). Consider the signal $y(t) = x_1(t) + x_1(-t)$. Using the time-reversal and conjugation properties of the CTFS, determine the coefficients for the CTFS of $y(t)$.

(c). Plot the signal $y(t)$ over $-1 \leq t \leq 1$. What type of symmetry do you expect? Can you explain what you see in terms of the symmetry properties of the CTFS?

$$a_k^{(1)} = \begin{cases} \frac{1}{2}, & k = \pm 1 \\ -\frac{1}{2}j, & k = 2 \\ \frac{1}{2}j, & k = -2 \\ 0, & others \end{cases}$$

$$a_k^{(2)} = \begin{cases} \frac{1}{2}, & k = \pm 1 \\ \frac{1}{2}j, & k = 2 \\ -\frac{1}{2}j, & k = -2 \\ 0, & others \end{cases}$$

$$a_k^{(y)} = a_k^{(1)} + a_k^{(2)} = \begin{cases} 1, & k = \pm 1 \\ 0, & others \end{cases}$$

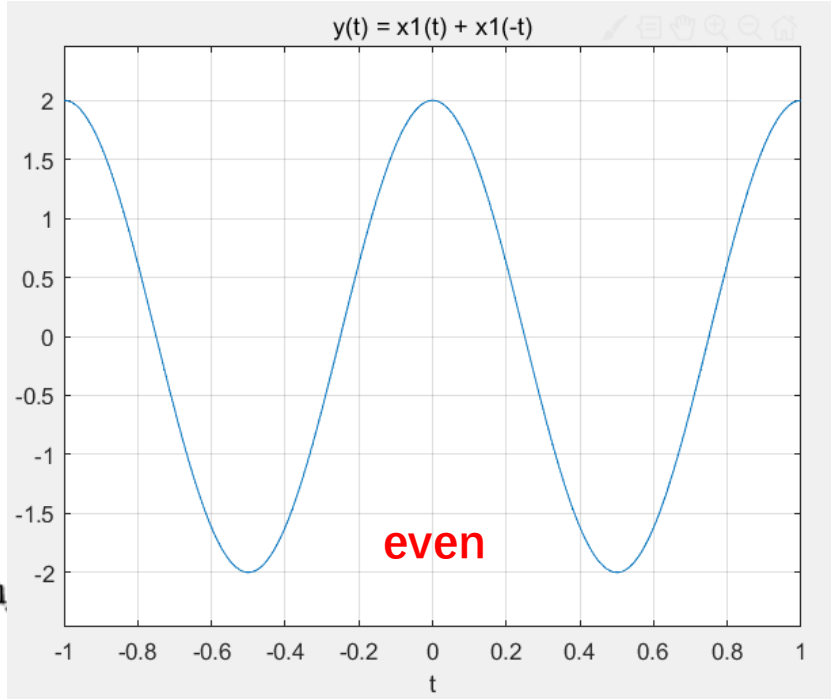
Background: Properties of CTFS

Showcase, From Lab book 3.6)

$$x_1(t) = \cos(\omega_0 t) + \sin(2\omega_0 t), \qquad \text{where } \omega_0 = 2\pi.$$

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(c). Plot the signal $y(t)$ over $-1 \leq t \leq 1$. What type of symmetry do you expect? Can you explain what you see in terms of the symmetry properties of the CTFS?



For y(t) = y(-t), is even.

Then FS coefficients are also even ($i.e., a_k^{(y)} = a_{-k}^{(y)}$)

How about odd ??

$$a_k^{(y)} = a_k^{(1)} + a_k^{(2)} = \begin{cases} 1, & k = \pm 1 \\ 0, & others \end{cases}$$

3, Synthesizing Continuous-Time signals with the Fourier Series

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Showcase, From Lab book 3.12) / Triangle Wave

The Fourier series synthesis equation for a continuous-time signal $x(t)$ with fundamental period T is given by Eq. (3.12). The Fourier series coefficients can, and generally do, have an infinite number of nonzero values. For instance, any signal with a discontinuity will have a Fourier series representation with an infinite number of nonzero coefficients, which is infeasible for numeric computations. The finite sum

$$x_N(t) = \sum_{k=-N}^{N} a_k e^{ik(2\pi/T)t} \tag{3.13}$$

is often a very good approximation for some relatively small integer N. Equation (3.13) is often called the truncated Fourier series representation of $x(t)$.

Truncated

$$x(t) = \sum_{k=-\infty}^{+\infty} a_k \phi_k(t) = \sum_{k=-\infty}^{+\infty} a_k e^{jk(2\pi/T)t}$$

3, Synthesizing Continuous-Time signals with the Fourier Series

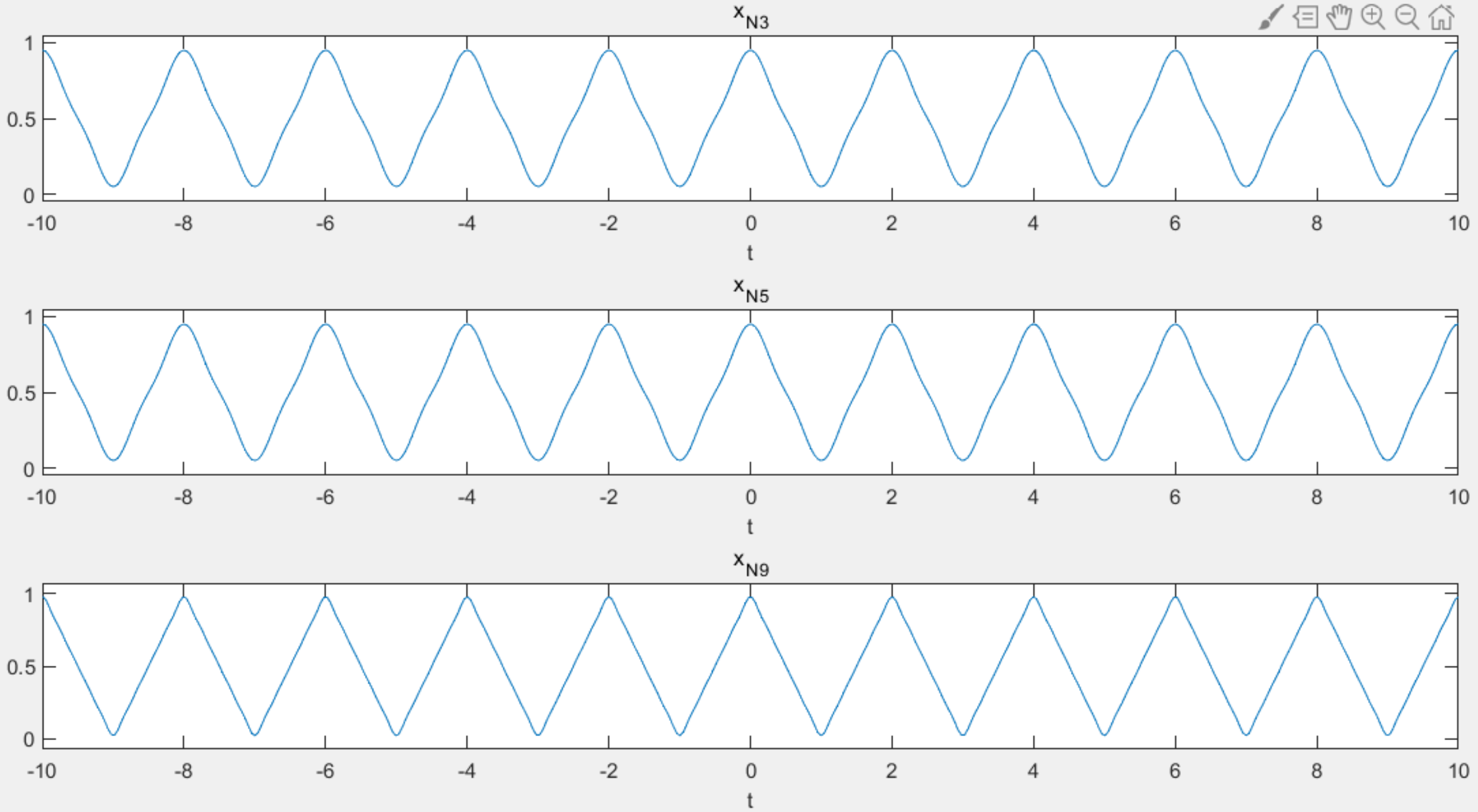
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Showcase, From Lab book 3.12) / Triangle Wave

(f). Create the symbolic expression for $x_N(t)$ for $N = 1$, 3, 5, and 9. Using `ezplot`, plot $x_N(t)$ on the interval $-1 \leq t \leq 1$. Use a different plot for each value of N.

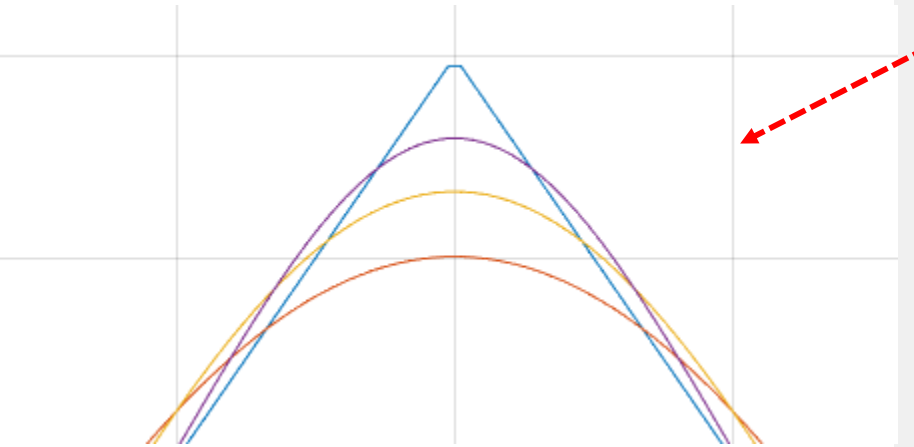
```
N = 3;
a_k = [a_k10_val(11-N:10), a0, a_k10_val(11:10+N)];
x_n3 = a_k(N+1); % a0, DC component
for k = 1:N
    % a_{k} and a_{-k}, nth harmonic component
    x_n3 = x_n3 + a_k(N+1-k) * exp(i * (-k) * (2 * pi / T) * t) +
a_k(N+1+k) * exp(i * k * (2 * pi / T) * t);
end

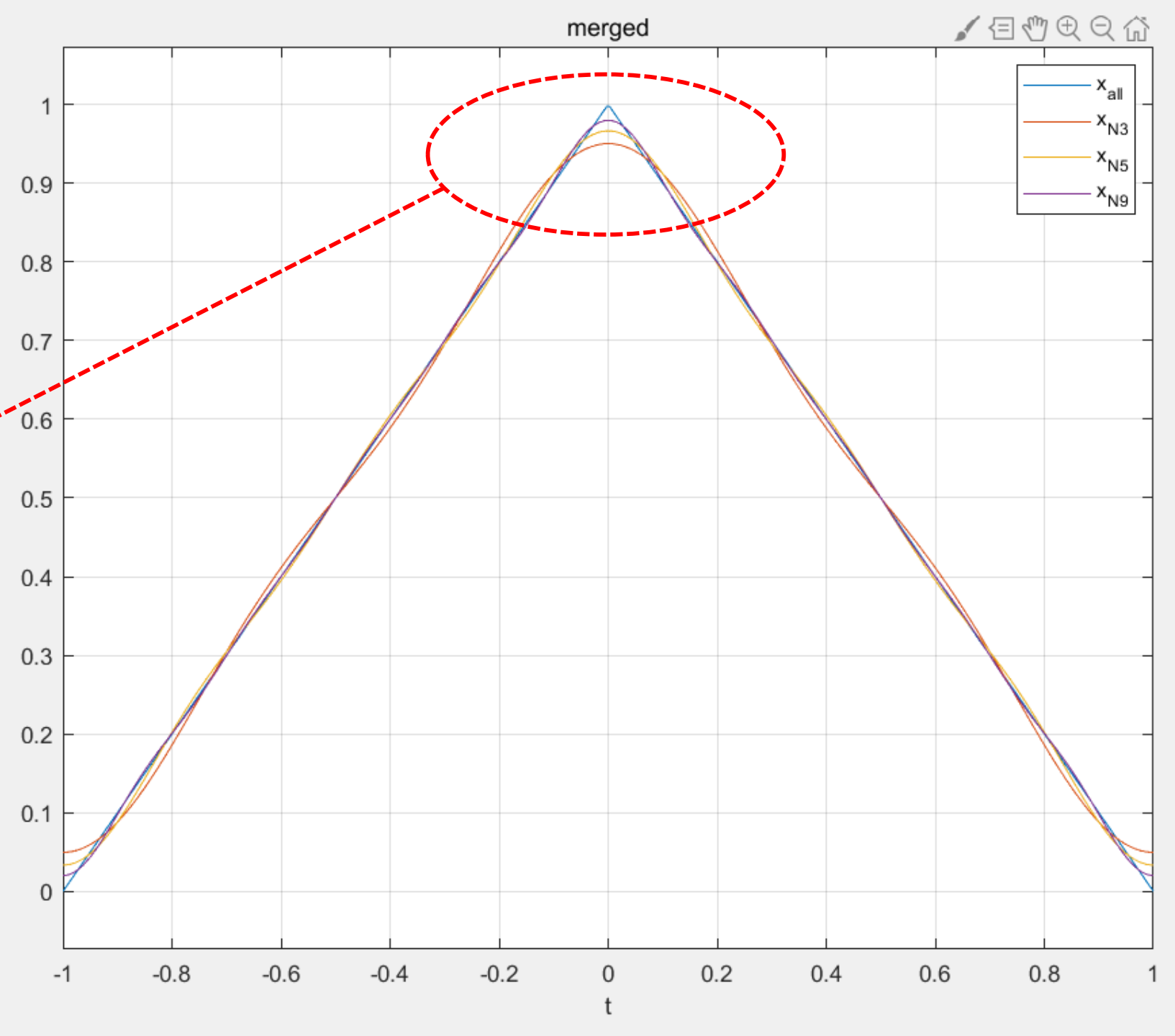
ezplot(x_n3, [-10, 10]);
title("x_{N3}");
```


x_{N3}
1
0.5
0
-10
-8
-6
-4
-2
0
2
4
6
8
10
t
x_{N5}
1
0.5
0
-10
-8
-6
-4
-2
0
2
4
6
8
10
t
x_{N9}
1
0.5
0
-10
-8
-6
-4
-2
0
2
4
6
8
10
t

3, Synthesizing Continuo

Showcase, From Lab book 3.12





3, Synthesizing Continuous-Time signals with the Fourier Series

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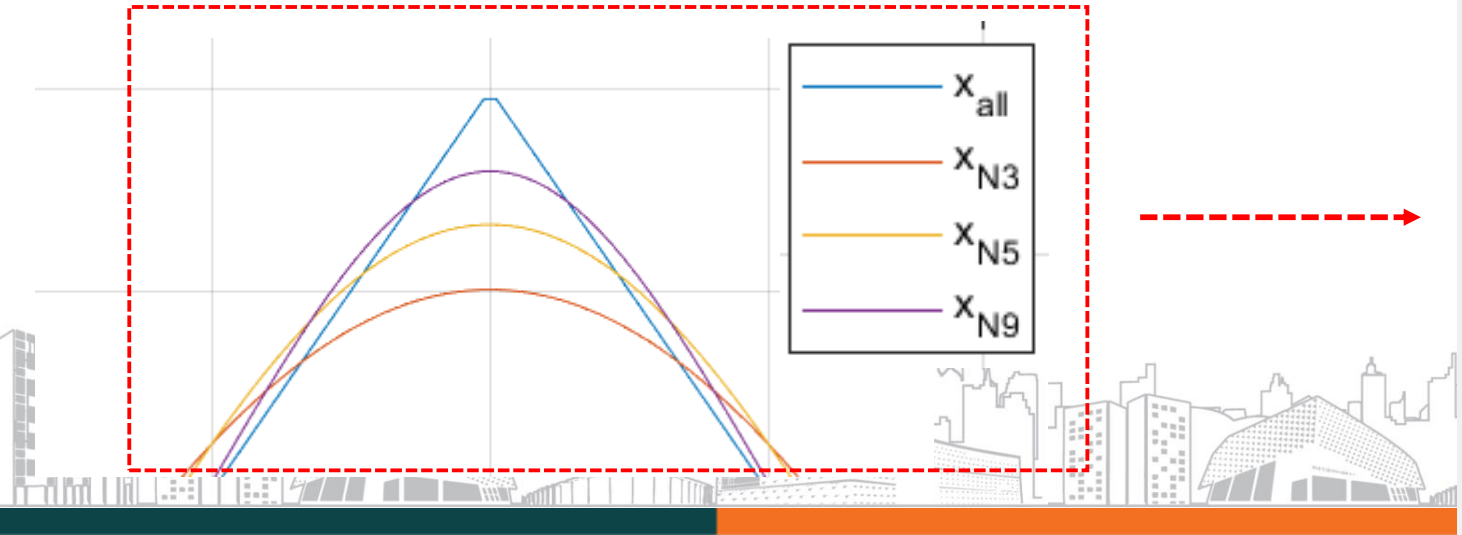
Showcase, From Lab book 3.12) / Triangle Wave

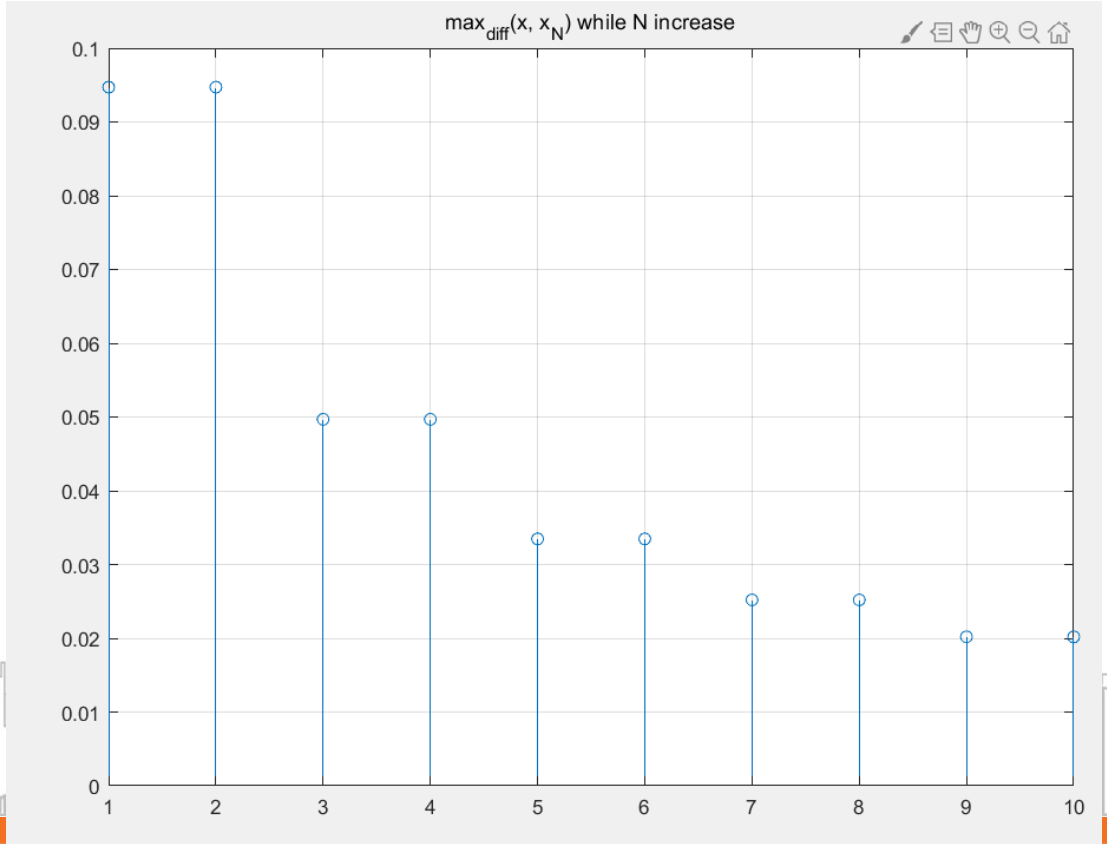
(f). Create the symbolic expression for $x_N(t)$ for $N = 1$, 3, 5, and 9. Using **ezplot**, plot $x_N(t)$ on the interval $-1 \leq t \leq 1$. Use a different plot for each value of N.

(g). How does $x_N(t)$ appear to converge for increasing N at $t = 0$? Does the maximum error

$$\max_{-1<t<1} |x(t) - x_N(t)|$$

appear to decrease as N increases? How does this behavior compare cated Fourier series approximation of the square wave?





4, The Fourier Representation of Square Waves

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From Lab Book 3.12, a) ~ d), as Lab Assignment

Consider a periodic square wave with fundamental period $T = 2$. Over the interval $-1 < t < 1$ the square wave is described by

$$x(t) = \begin{cases} 1, & |t| \leq \frac{1}{2}, \\ 0, & |t| > \frac{1}{2}, \end{cases}$$

as illustrated in Figure 3.6. In this exercise, you will analyze the Fourier series representation of the square wave, focusing upon the neighborhood of the discontinuities in the square wave.

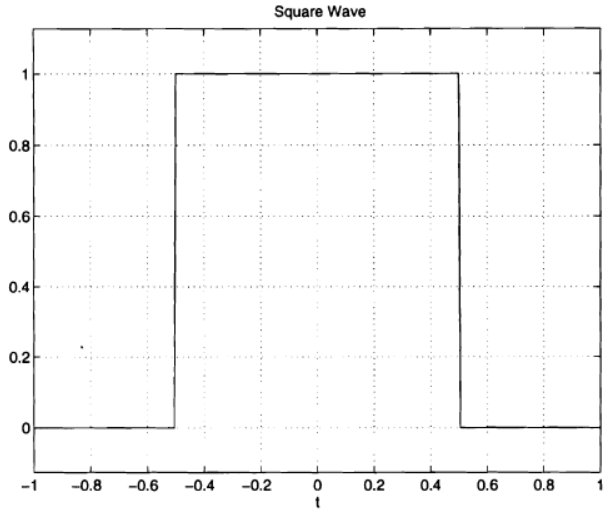


Figure 3.6. A single period of the square wave.

4, The Fourier Representation of Square Waves

From Lab Book 3.12, a) ~ d), as Lab Assignment

Ideas： by using symbolic expression and 'int' (积分运算), calculate CTFS a_k

* Note： if the 'numeric(…)' function error， change to 'double(…)' function instead

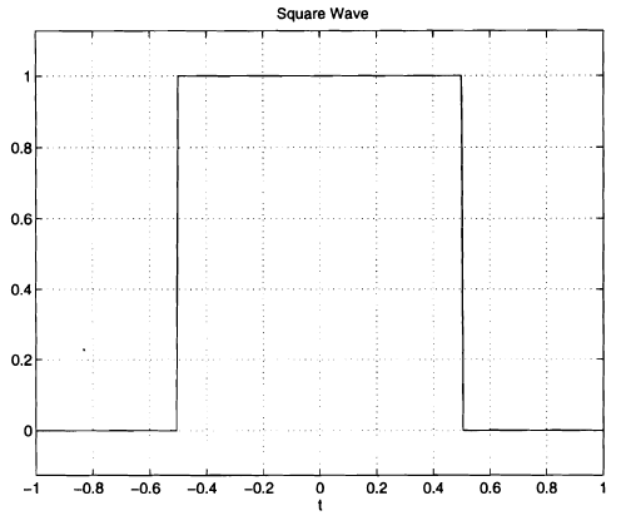


Figure 3.6. A single period of the square wave.

(a). Using `int`, create a symbolic expression `a` which contains the Fourier series coefficients of the square wave for each value of k. The symbolic expression will be a function of k, e.g., a_5 is given by `numeric(subs(a,5,'k'))`. Simplify this expression as much as possible. Use `numeric` and `stem` to plot the Fourier series coefficients for $-10 \leq k \leq 10$. Note: Depending upon how you simplify your symbolic expression for `a`, MATLAB may not be able to evaluate `a` for $k = 0$ since MATLAB cannot evaluate expressions like `sin(k)/k` at $k = 0$. In this case, calculate a_0 separately.

4, The Fourier Representation of Square Waves

From Lab Book 3.12, a) ~ d), as Lab Assignment

Ideas：signal synthesize analysis, by CTFS coefficients a_k

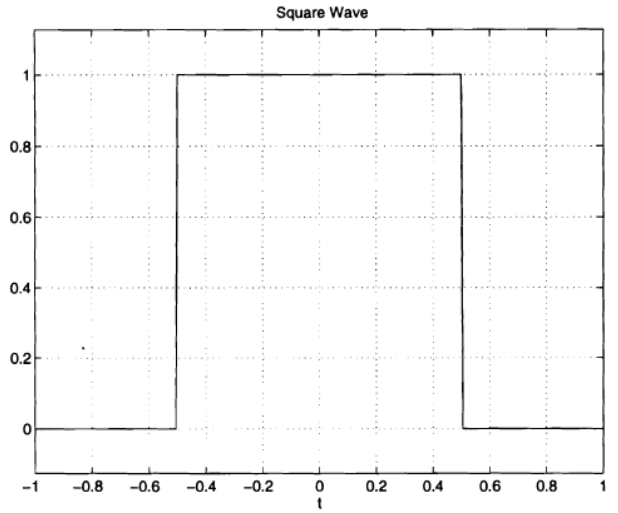


Figure 3.6. A single period of the square wave.

(b). Create the symbolic expression for $x_N(t)$ for $N = 1, 3, 5$, and 9. Using `ezplot`, plot $x_N(t)$ on the interval $-1 \leq t \leq 1$. Plot all four signals on the same figure using `hold`.

(c). What is the value of $x_N(t)$ at $t = \pm 1/2$? Does the value change as N increases?

(d). Without explicitly evaluating $x_N(t)$, estimate the value of the overshoot error for each value of N. Does the overshoot error decrease as N increases? How do you expect this value to change as $N \longrightarrow \infty$? (If you have the time and resources, you might plot $x_N(t)$ for $N > 9$.)

Lab3 / Task

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3.5:

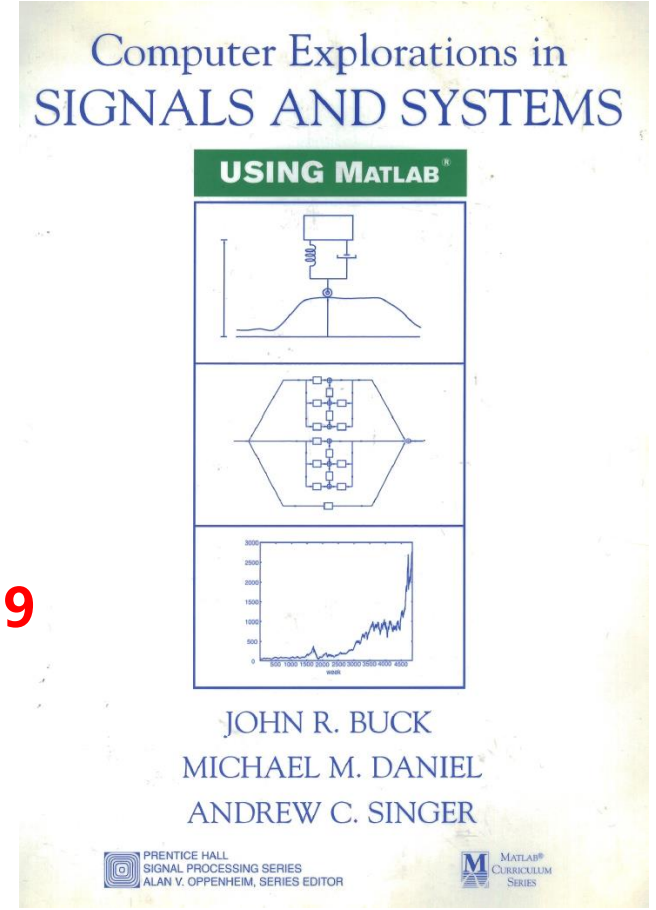
- a) ~ c)
- d) ~ h) for x3
- Except Gibb ‘s problem of h)

3.12:

- a) ~ d), for square wave

Deadline: 19 April, Sunday 23:59

Submit to Blackboard



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THANK YOU

